

OMAR NAGY MOHAMMED

(+20) 101 433 7548 | Hadayek El Ahram, Giza, Egypt | omar.badran.work@gmail.com | github.com/bradocola | linkedin.com/in/omar-nagy-work

SUMMARY

Third-year Computer Engineering student at Cairo University with hands-on experience across the embedded stack, from VHDL-based processor design to ESP32/STM32 firmware and real-time, sensor-driven control systems. Built and competed with an autonomous vehicle system (4th of 48 teams, CUERT) and a custom 5-stage pipelined RISC-like processor with data forwarding and branch prediction.

SKILLS

Languages & Tools: C, C++, VHDL, Verilog, Assembly, Python, JavaScript, Git, ModelSim

Platforms: ESP32, STM32

Foundations: Object-Oriented Programming, Data Structures & Algorithms

Soft Skills: Problem Solving, Time Management & Prioritization

WORK EXPERIENCE

Mobile Developer Intern — Park St. Developments

June 2026 - August 2026

Al Sheikh Zayed City, Giza

- Spent 8 of a 10-week internship on mobile application development tasks, contributing to features under the guidance of the mobile team.
- Gained 2 weeks of cross-functional exposure to IT infrastructure, covering CCNA fundamentals including NAT configuration, router/switch data flow, and firewall security policies.

PROJECTS

Automated Car — CUERT Contest

2025

Using C

- Implemented deterministic navigation and sensor-based lane-detection logic in C for automated zone routing and timed obstacle stops, placing 4th out of 48 teams in the CUERT automated vehicle competition.

CUBIX

2026

Using ESP32 & C++

- Collaborated on an 11-member team to build a connected system linking a hardware cube with desktop and mobile applications.
- Developed firmware and hardware logic on ESP32 to achieve real-time state detection and multi-device synchronization.

Computer Architecture Project

2025

Using VHDL

- Implemented a custom 5-stage pipelined processor in VHDL to execute a RISC-like instruction set, including data forwarding and static branch prediction.

Game Boy Microprocessor Project

2025

Using Assembly

- Built a Game Boy from basic components with custom-developed software.

EDUCATION

Faculty of Engineering, Cairo University

2023 - 2028 (Expected)

B.Sc. in Computer Engineering

CERTIFICATIONS

CS50x & CS50P: Computer Science & Python Programming, Harvard University

LANGUAGES

Arabic | English